Effect of diet and exercise intervention on the growth of prostate epithelial cells.

Epidemiological studies suggest a positive association between nutrient intake, hyperinsulinemia and risk of Benign prostatic hyperplasis (BPH). This study tests the hypothesis that a low-fat, high-fiber diet and daily exercise would lower serum insulin and reduce the growth of serum-stimulated primary prostate epithelial cells in culture. Serum samples were obtained from eight overweight men before and after the Pritikin residential, 2-week diet and exercise intervention and from seven men who were long-term followers of the low-fat, high-fiber diet and regular exercise lifestyle. The serum was used to stimulate primary prostate epithelial cells in culture. Growth was measured after 48 and 96 h and apoptosis after 96 h. At 48 h there was no significant difference in growth within the Pre, 2-week or Long-Term groups. At 96 h growth was significantly reduced in the 2-week (13%) and in the Long-Term (14%) groups compared to the Pre data. At 96 h, apoptosis was not significantly different among the three groups. Fasting insulin was reduced by 30% in the 2-week group and by 52% in the Long-Term group compared to the Pre data. Testosterone was unchanged in the 2-week group. The results of this study indicate that a low-fat, high-fiber diet and daily exercise lowers insulin and reduces growth of prostate primary epithelial cells and suggests that lifestyle may be an important factor in the development or progression of BPH. Future prospective trials should address the effects of this lifestyle modification on BPH symptomatology and progression.